MEDICINE-STOCK-CHECKER

Count the days, then look sideways before you look up

1. Introduction

Medicine-Stock-Checker is a proposed medicine inventory monitoring and early warning system designed for Primary Health Centres (PHCs). The project focuses on identifying potential medicine shortages before they affect patients and helping healthcare staff take timely action.

In many situations, one health centre may run out of an essential medicine while another nearby centre has more stock than it currently needs. When shortages are identified late and replenishment depends on periodic district-level orders, patients may experience unnecessary delays in receiving medicines.

Medicine-Stock-Checker aims to address this problem by forecasting stock-out dates, identifying nearby centres with available surplus stock, and recommending transfers before requesting additional supplies from the district.

2. Problem Statement

Medicine Stock-Out Early Warning for Primary Health Centres

Primary Health Centres need a reliable way to monitor essential medicine availability and anticipate shortages. Looking only at the current stock quantity may not reveal how quickly a medicine is being consumed or how soon it will run out.

At the same time, surplus stock at nearby centres may remain unused while another centre faces an urgent shortage. A system that considers both consumption trends and stock availability across centres can help staff make better replenishment decisions.

3. Proposed Solution

Medicine-Stock-Checker combines stock monitoring, consumption-based forecasting, and local medicine redistribution in one system.

The system estimates how many days of stock remain at each centre and identifies medicines that may run out soon. It then checks whether nearby centres can safely share surplus stock. If the shortage cannot be fully covered through redistribution, the remaining requirement is calculated for a district-level replenishment request.

This approach aims to support earlier decisions, better coordination, and more effective use of available medicine stocks.

4. Objectives

- Monitor essential medicine stock across multiple Primary Health Centres.
- Estimate the number of days remaining before a medicine runs out.

- Identify potential shortages early enough for action.
- Calculate the shortage at each centre and the safe surplus available elsewhere.
- Recommend suitable transfers between nearby centres.
- Calculate the remaining quantity required from the district.
- Provide a clear dashboard for monitoring stock levels and urgent requirements.

5. Key Features

1. **Medicine Stock Monitoring**

Maintain inventory records for different medicines at participating health centres, including current quantities and recent consumption.

2. **Stock-Out Forecasting**

Estimate the remaining days of stock using recent average daily consumption. Use the estimate to calculate a projected stock-out date.

3. **Early Warning Alerts**

Classify medicine availability into Critical, Warning, and Safe categories. Alert thresholds should account for expected replenishment time rather than stock quantity alone.

4. **Surplus Identification**

Estimate how much stock a centre can share after accounting for its own expected consumption and required safety stock.

5. **Nearby Centre Matching**

Recommend transfers by prioritizing urgent shortages and considering distance, available surplus, and transfer feasibility.

6. **District Replenishment Support**

Calculate the shortage that remains after suitable local transfers and prepare the information needed for a district replenishment request.

7. **Transfer Tracking**

Track proposed transfers through Suggested, Approved, In Transit, and Received stages. Update inventory when receipt is confirmed.

8. **Inventory Dashboard**

Present stock quantities, forecasted run-out dates, urgency levels, transfer recommendations, and outstanding district requirements in one view.

6. Working Methodology

The proposed system follows these steps:

1. Collect stock quantities and recent consumption records from each health centre.
2. Calculate average daily consumption using an appropriate recent observation period.
3. Estimate days remaining and the projected stock-out date.
4. Identify centres requiring urgent replenishment.
5. Calculate the shortage at each centre and the safe surplus available at potential donor centres.
6. Match shortages with suitable nearby donors, starting with the most urgent requirements.
7. Reserve allocated quantities so that the same surplus is not promised to multiple centres.
8. Calculate any remaining shortage and escalate it for district replenishment.
9. Track transfer approval and receipt, updating stock records when the medicine arrives.

If consumption is zero or records are outdated, the system should flag the situation for review instead of producing misleading forecasts.

7. Example Scenario

Consider two Primary Health Centres that maintain paracetamol syrup stocks.

Particulars	**Centre A**	**Centre C**
Current stock	60 bottles	200 bottles
Average daily consumption	10 bottles	2 bottles
Estimated days remaining	6 days	100 days
Distance between centres	9 km	9 km

Assume that the target stock cover is 30 days.

Centre A requires 300 bottles for 30 days. Since it currently has 60 bottles, its shortage is 240 bottles.

Centre C requires 60 bottles for 30 days. With 200 bottles in stock, it has a potential surplus of 140 bottles.

The system can therefore recommend transferring up to 140 bottles from Centre C to Centre A, subject to authorization and operational checks. The remaining shortage of 100 bottles can then be included in a district replenishment request.

This example illustrates how consumption-based forecasting and local redistribution can work together to address medicine shortages.

8. Technology Stack

The initial prototype is planned using the following technologies:

- **HTML:** Structure of the web application.
- **CSS:** Layout, styling, and dashboard design.
- **JavaScript:** Forecasting calculations, stock matching, and interface interactions.
- **Firebase Firestore:** Storage of centre, medicine, and inventory records.
- **Firebase Authentication:** User authentication and access management.

AI-assisted development tools may be used to support implementation, debugging, and testing.

9. Data Security and Responsible Use

The prototype will initially use simulated health-centre and inventory data. Real patient information is not required for the demonstration.

Database access should be restricted through appropriate authentication and security rules. Medicine transfers should require approval from authorized personnel, and inventory records should be updated only after confirmed receipt.

The system is intended to support inventory planning. It does not provide medical advice or replace the judgment of healthcare professionals.

10. Expected Outcomes

Medicine-Stock-Checker aims to improve medicine availability by identifying shortages before stock runs out and making existing surplus stock more visible across nearby centres.

The expected outcomes include earlier shortage warnings, better coordination between health centres, more informed district replenishment requests, and reduced avoidable delays in responding to medicine requirements.

Actual improvements would need to be evaluated using reliable inventory data and real-world testing.

11. Future Enhancements

Possible future improvements include:

- Warnings for outdated stock records.
- Detection of unusual increases in medicine consumption.
- SMS or email alerts.
- Automated district indent request generation.
- A Tamil-language interface.

- Interactive maps showing nearby health centres.
- Improved forecasts using historical consumption data.

12. Project Status

Current status: Initial planning and development.

This project is being developed for Sakthi Hack Fest '26 under the Sustainable Development Goals domain. Features will be described as completed only after implementation and testing.

13. Team Details

Team Name: Beginner's

Hackathon: Sakthi Hack Fest '26

Team Members:

- Pradeepa S — Team Leader
- Gokulnath S
- Maha S
- Nithyadharsini B

Conclusion

Medicine-Stock-Checker proposes a practical way to move from reacting to medicine shortages towards anticipating them. By estimating how quickly stock is falling, checking for safe surplus at nearby centres, and escalating only the remaining requirement, the system aims to help healthcare staff make timely and